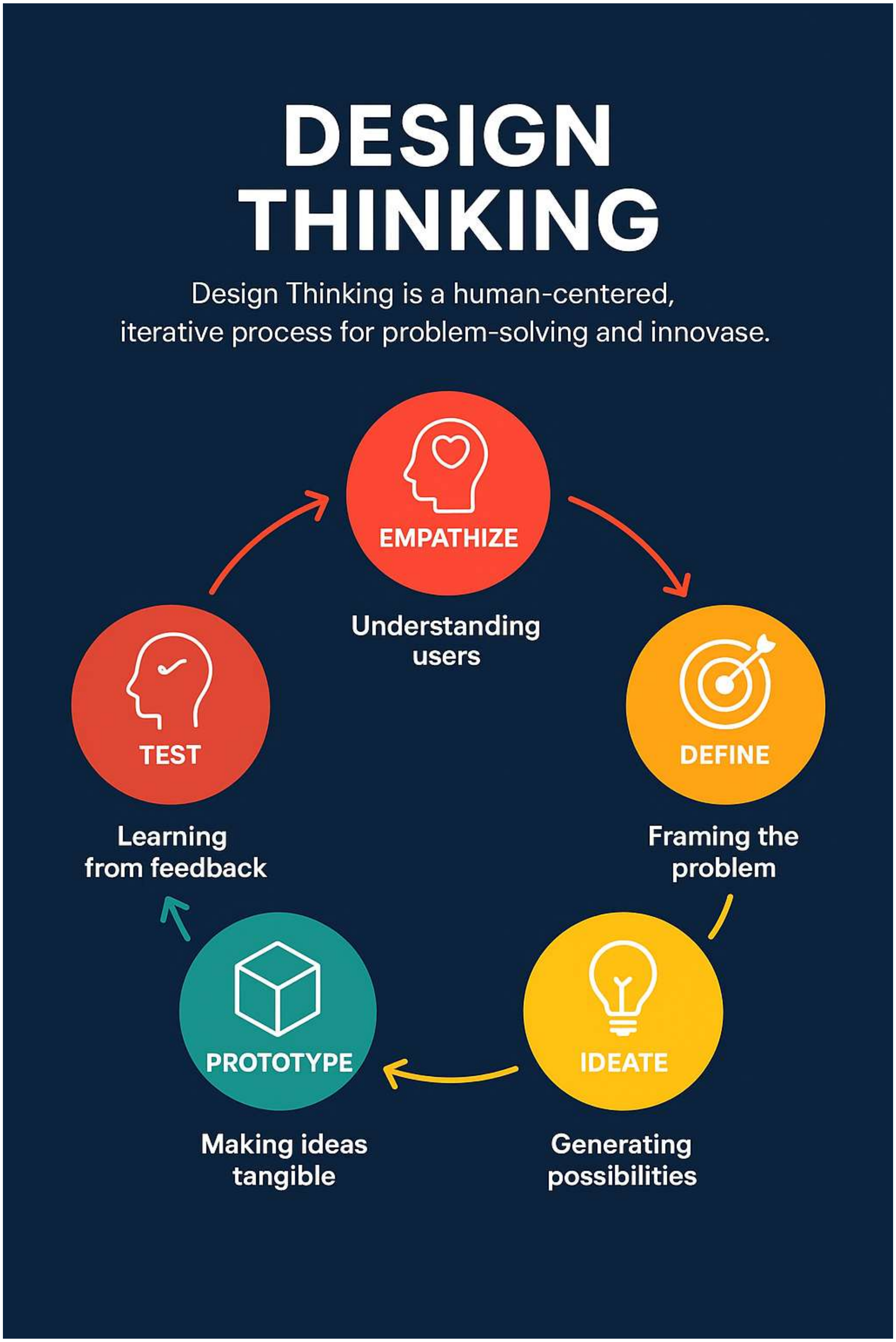
UNIT-2 Design Thinking Framework



A design thinking framework is a human-centered problem-solving process with distinct, often iterative phases, most commonly including Empathize, Define, Ideate, Prototype, and Test. This flexible framework emphasises understanding users, creatively exploring solutions through ideation, developing tangible prototypes, and testing those prototypes with real users to gather feedback for refinement and iteration. The process is not strictly linear, allowing teams to revisit earlier stages as new insights emerge, ensuring the final solution is innovative, user-friendly, and effective.

The Design Thinking Process: Empathise, Define, Ideate, Prototype, Test

1. **Empathise:**
    - **Focus:** Understand the needs, motivations, and challenges of your users.
    - **Actions:** Conduct user research, interviews, and observations to gain deep insights into their experiences and perspectives.
2. **Define:**
    - **Focus:** Synthesize the insights from the empathize phase to form a clear and concise problem statement.
    - **Actions:** Reframe the challenge from a user-centric point of view, ensuring the problem is well-understood before rushing to a solution.
3. **Ideate:**
    - **Focus:** Generate a wide range of creative solutions to the defined problem.
    - **Actions:** Engage in brainstorming and idea generation, encouraging a judgment-free environment to allow for bold and innovative ideas to surface.
4. **Prototype:**
    - **Focus:** Turn abstract ideas into tangible models or low-fidelity versions of potential solutions.
    - **Actions:** Create mockups, sketches, or basic versions of the solution to visualize and explore concepts.
5. **Test:**
    - **Focus:** Evaluate the prototypes with actual users to gather feedback and identify areas for improvement.

- **Actions:** Test prototypes in real-world situations and use the feedback to refine the solution, potentially returning to earlier phases for iteration.

Key Characteristics of the Framework

- **User-Centric:**

The entire process is centered around understanding and meeting user needs.

- **Iterative:**

The phases are not always followed in a linear order; you may need to return to previous stages to incorporate new learnings and refine the solution.

- **Collaborative:**

It encourages teams from diverse backgrounds to work together, fostering knowledge exchange and innovation.

- **Action-Oriented:**

The focus on prototyping and testing ensures that ideas are explored through hands-on experimentation.

Techniques to do Empathize Step

- **Empathy Mapping**
  Captures what users *say, think, feel, and do*.
- **Observation & Shadowing**
  Watch users in real situations → find hidden pain points.
- **Interviews & Storytelling**
  Hear personal experiences to uncover needs.
- **Surveys & Feedback Tools**
  Collect structured insights at scale.
- **Customer Journey Mapping**
  Map user's step-by-step interaction → spot friction points.
- **5 Whys + Root Cause Analysis**
  Dig deeper to find underlying issues.
- **Affinity Diagramming**
  Cluster ideas/pain points into themes.
- **Personas**
  Create user profiles to focus on real needs.

Research Methods

Qualitative Research (Deep Understanding)

- **Ethnographic Studies** – immersion in user's environment.
- **In-depth Interviews** – to explore attitudes & emotions.
- **Focus Groups** – collective discussion for diverse insights.
- **Case Studies** – learn from specific examples.

Quantitative Research (Validation & Scale)

- **Surveys & Polls** – measure frequency of problems.
- **Usage Data Analysis** – identify behavior trends.
- **A/B Testing & Experiments** – test user responses.

Mixed/Iterative Methods

- Combine **observation + data** for accuracy.
- Example: *Shadow users → design survey → validate insights.*

<u>Techniques to do Define Step</u>

1. Synthesize Research Findings

- Review data collected during the **Empathy stage** (interviews, observations, user journeys).
- Identify **patterns, themes, and insights** about user needs and pain points.

2. Create User Personas

- Develop fictional but realistic characters representing key user groups.
- Capture details like demographics, motivations, challenges, and goals.
- This ensures the design focuses on **real people's needs**.

3. Map the User Journey

- Visualize the steps users take when interacting with a product/service.
- Highlight **friction points** and **moments of opportunity**.

4. Frame the Problem

- Craft a **Problem Statement** that is:
  - **Human-centered** (focuses on the user, not the company).
  - **Actionable** (guides idea generation).
  - **Focused** (specific but not limiting creativity).

Example: Instead of *“We need to improve our app”*, write *“How might we help busy college students manage their time better through our app?”*

5. Define the “Point of View (POV)”

- Combine **user + need + insight** into one concise statement.
- Example: *“A busy student who needs an easy way to organize tasks because traditional planners feel overwhelming.”*

✅ **Key Goal of Define step**: Narrow down to a **clear design challenge** that sparks creative solutions in the next step (Ideation).

<u>Techniques to do Ideate Step</u>

1. Set the Stage

- Begin with the problem statement or *How Might We…?* question defined in the previous step.
- **How Might We” (HMW)** is a **framing technique** used in the Define step of Design Thinking. It transforms user needs, pain points, or insights into open-ended, opportunity-driven questions that inspire creativity during Ideation.

▪ **What is *How Might We (HMW)*?**

- A way to reframe challenges as design opportunities.
- Structured as:
  “How might we [do something] for [user] in order to [achieve outcome]?”
- Keeps the tone positive, open, and solution-oriented.

▪ **Why use *How Might We*?**

- Encourages multiple solutions (not one “correct” answer).
- Bridges the gap between Define (problem clarity) and Ideate (solution generation).
- Ensures the design challenge is human-centered.

▪ **Examples of *How Might We* Statements**

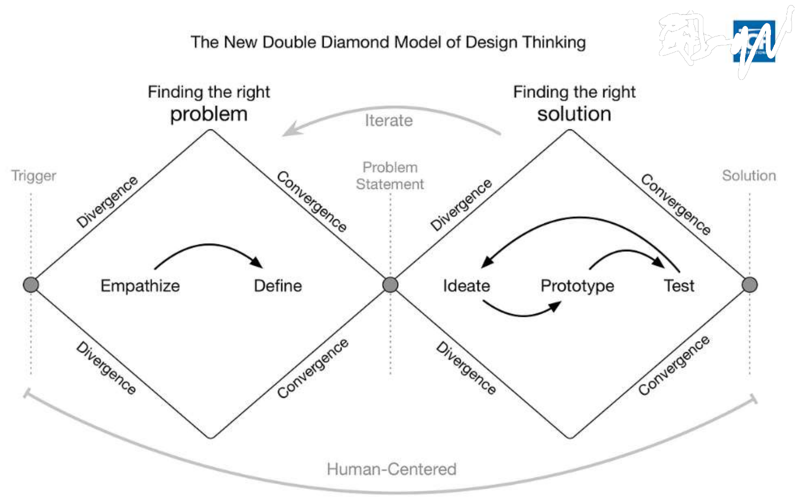
- From healthcare:
- *How might we help elderly patients remember to take their medication on time?*
- From education:
- *How might we make online learning more engaging for high school students?*
- From transportation:
- *How might we reduce stress for daily commuters during peak hours?*

2. Divergent Thinking (Idea Generation)

- Use creativity techniques to brainstorm multiple ideas:
  - **Brainstorming** – quick, free-flowing ideas in groups.
  - **Brainwriting** – individuals write ideas silently, then share.
  - **SCAMPER** – Substitute, Combine, Adapt, Modify, Put to another use, Eliminate, Reverse.
  - **Mind Mapping** – visually connecting ideas around a central concept.
  - **Worst Possible Idea** – explore "bad" ideas to spark new thinking.

3. Convergent Thinking (Narrowing Down)

- After idea generation, cluster and evaluate ideas.
- Use methods like:
  - **Dot voting** – team members vote on the most promising ideas.
  - **Impact vs. Feasibility Matrix** – map ideas based on value and practicality.
  - **Prioritization grids** – decide which to test first



4. Selection of Ideas

- Select a few high-potential ideas (not just the "safe" ones).
- Combine or refine ideas if necessary.
- Carry them forward into prototyping

✅ **Key Goal of Ideation**: Move beyond obvious solutions and open up to innovative, user-centered possibilities.

<u>Techniques to do Prototype Step</u>

1. Select Promising Ideas

- From the **Ideation step**, pick a few high-potential solutions.
- Focus on the ones that best address the user's needs and are feasible.

2. Build Low-Fidelity Prototypes

- Start simple and inexpensive — sketches, paper mock-ups, role-playing, storyboards, or cardboard models.
- Purpose: **communicate and test the concept quickly**, not to build the final product.

3. Iterate to Medium/High-Fidelity

- Develop more detailed prototypes such as:
  - **Wireframes** or clickable digital prototypes.
  - **3D models** or functional mock-ups.
  - **Service blueprints** for service design.
- Increase fidelity step by step, based on feedback.

4. Keep it User-Focused

- Always design prototypes with the **end-user in mind**.
- Build enough for users to interact with and give feedback.

5. Test & Refine

- Share prototypes with team members or target users.
- Collect feedback: What works? What confuses? What excites?
- Refine or pivot the prototype based on insights.

✅ **Key Goal of Prototype Step**: To **bring ideas to life quickly**, explore different directions, and prepare for **real-world testing** in the next stage.

<u>Techniques to do Test Step</u>

1. Prepare for Testing

- Define what you want to learn from the test.
- Decide the type of feedback needed: usability, functionality, desirability, or overall experience.
- Select the target users (same personas identified earlier).

2. Engage Users with Prototypes

- Let users **interact with prototypes** in real or simulated scenarios.
- Observe how they use it without giving too many instructions (to see natural behavior).
- Ask them to "think aloud" as they use it to reveal their thought process.

3. Collect Feedback

- Use multiple methods:
  - Direct observation.
  - Interviews or focus groups.

    - Surveys, rating scales, or feedback forms.
- Focus on *why* users behave a certain way, not just *what* they do.

4. Analyze Results

- Look for **patterns in feedback**: What worked? What caused confusion? What delighted users?
- Identify both strengths and weaknesses of the prototype.

5. Refine & Iterate

- Use insights to **modify prototypes** or even revisit earlier stages (ideation, define, empathize).
- Repeat the cycle until the solution is user-centered, functional, and desirable.

✅ **Key Goal of Test Step**: To validate whether the solution truly meets **user needs**, and refine it until it's practical, valuable, and user-approved.

Case Study: Embrace Infant Warmer

Background

- Every year, 15 million premature and underweight babies are born globally.
- In developing countries like India, many die due to hypothermia (inability to maintain body temperature).
- Standard solution: Incubators (~$20,000 each) → too expensive, require electricity, not feasible in rural areas.
- Mothers in villages often resort to unsafe methods (fires, hot water bottles, cloth wrapping).
- Need: A low-cost, portable, reliable, and culturally acceptable solution.

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1. Empathize

- Student innovators (Stanford d.school) visited rural India.
- Interacted with mothers, doctors, and health workers.
- Insight:
  - Premature babies often born at home or in small clinics with no incubators.
  - Mothers deeply desired a safe, affordable method to save their babies.

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2. Define

- Challenge Statement:
  How might we design a baby-warming solution that is affordable, portable, electricity-independent, and easy to use in rural India?

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3. Ideate

- Brainstormed around low-tech + high-impact ideas.
- Explored alternatives: hot-water bottles, heating lamps, blankets.
- Final concept → portable infant warmer with thermal regulation.

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4. Prototype

- Developed a sleeping-bag style wrap for babies.
- Inserted a phase-change material pouch that could be heated with hot water/other sources.
- Maintains 37°C for 4–6 hours – the critical range for infant survival.
- Designed to be reusable, safe, and affordable.

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5. Test

- Field trials in rural hospitals & homes.
- Feedback:
  - Easy to use for mothers & health workers.
  - Babies maintained healthy body temperature.
  - Adaptable in both clinics and homes.
- Iterated design for durability and cultural fit.

Outcome & Impact

- Product: Embrace Infant Warmer (resembles a baby sleeping bag).
- Cost: ~$25 compared to $20,000 incubator.
- Reach: Used in 20+ countries, saving 300,000+ infants.
- Recognition: Awarded by World Economic Forum, Skoll Foundation, and TED.
- Impact: Empathy-driven, frugal innovation that redefined affordable healthcare design.

Link

Bing Videos

CASE STUDY: GE VSCAN PORTABLE ULTRASOUND THROUGH DESIGN THINKING

	Case Study: GE Vscan Portable Ultrasound through Design Thinking Background

The GE Vscan is a handheld, pocket-sized ultrasound device developed by GE Healthcare. It aimed to revolutionize medical diagnostics by making imaging portable, affordable, and accessible in diverse healthcare settings—from hospitals to rural clinics.

Applying Design Thinking Stages

1. Empathize

User Needs:

- Doctors in remote and resource-constrained areas needed fast diagnostic tools.
- Patients required quick, accurate, and affordable imaging without depending on large hospitals.

Pain Points:

- Conventional ultrasound machines were bulky, expensive, and restricted to specialized centres.
- Patients in rural/remote locations faced delays and high costs traveling for diagnostics.
- Empathy Insight: Healthcare workers needed a portable, reliable, and easy-to-use diagnostic tool to make immediate decisions

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2. **Define**

Problem Statement:

- "How might we design a portable, easy-to-use, and affordable ultrasound device that enables doctors to make fast and accurate diagnostic decisions anywhere?"

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3. **Ideate**

Ideas Explored:

- Shrink the size of ultrasound technology to a handheld device.
- Simple user interface with minimal training requirements.
- Wireless/USB connectivity to store and share results easily.
- Affordable pricing model to make it accessible for emerging markets.
- Battery-powered for use in off-grid locations.

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4. **Prototype**

- GE created multiple prototypes of a pocket-sized ultrasound scanner.
- Doctors tested prototypes in both urban hospitals and rural clinics.
- Feedback: Needed clearer images, intuitive navigation, and rugged design for field use.

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5. **Test**

- Iterative testing with healthcare providers globally:
- In rural India and Africa → tested for accessibility and ruggedness.
- In emergency rooms → tested for speed and accuracy.
- Adjustments were made for battery life, image resolution, and durability.
- Final product: GE Vscan, a handheld ultrasound scanner the size of a smartphone.

Design Thinking Principles in Action

- Human-Centered: Designed around doctors' and patients' needs in underserved areas.
- Simplicity: Easy interface, no complex training required.
- Accessibility: Portable, affordable, and usable in any setting.
- Iterative Innovation: Refined through multiple rounds of global testing.
- Scalability: Now used in over 100 countries by clinicians, paramedics, and rural healthcare workers.

Key Lessons

- Empathy with underserved users (rural patients, frontline doctors) drives breakthrough innovations.
- Simplifying advanced technology makes it more powerful and impactful.
- Iterative prototyping and testing in real environments ensures practical success.
- Healthcare innovations must balance cost, usability, and accuracy.
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Link:(50) GE Vscan Air Review | MSK Handheld Ultrasound Tested & Compared - YouTube

Case Study: Tally Software

Background

- **Problem Context**: In India, small and medium businesses (SMBs) traditionally struggled with **manual bookkeeping**. Accounting was **complex, time-consuming, error-prone**, and required trained accountants.
- **Need**: A simple, affordable, and localized solution to manage accounts, inventory, payroll, and taxes.
- **Company**: Tally Solutions Pvt. Ltd. → Started in 1986, became a pioneer in accounting software for SMBs.

Design Thinking Journey

1. Empathize

- Spent time with **small shop owners, accountants, and entrepreneurs**.
- Observed their pain points:
    - Complicated government tax compliance.
    - Lack of accounting knowledge.
    - Desire for quick, reliable financial reports.
    - Need for **software in local languages**.

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2. Define

Problem Statement:

How might we create a simple, reliable, and affordable software that enables Indian businesses (even those with no deep accounting knowledge) to manage accounts, inventory, and compliance easily?

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3. Ideate

Brainstormed solutions around:

- **Automation** → auto ledger & voucher entries.
- **Simplicity** → intuitive interface, keyboard shortcuts, local language support.
- **Scalability** → one product usable by both small kirana shops and large enterprises.
- **Compliance** → automated GST, VAT, payroll, and statutory updates.

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4. Prototype

- Early versions (Tally 4.5, 6.3, etc.) tested with **local businesses**.
- Focus on **speed, low system requirements, and offline usability**.

- Added **multilingual support** and features like **inventory + financial accounting integration**.

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5. Test
Pilots with accountants, CA firms, retailers, and SMEs.

- Feedback refined:
  - Simple **keyboard-driven commands** (faster than GUI clicks).
  - **Real-time reporting** → immediate profit/loss visibility.
  - Continuous improvement with Tally ERP and later **Tally Prime**.

Outcome & Impact

- **Adoption**: Over **2 million businesses** use Tally across 100+ countries.
- **Innovation**:
  - Simplified GST compliance in India.
  - Localized features for Indian businesses.
- **Impact**: Empowered **SMBs with digital tools** for growth.
- **Design Thinking Lesson**: Understanding end-user struggles (small traders, accountants) → building **simple yet powerful software** → iterating with user feedback → scaling to millions.

Case Study: Lucky Iron Fish through

Background

- **Problem:** Severe **iron deficiency & anaemia** in rural Cambodia, especially among women and children.
- Pills/supplements were expensive, had side effects, and were not consistently used.
- Needed: **affordable, culturally acceptable, sustainable solution**.

Design Thinking Stages

1. Empathize

- Field research in Cambodian villages.
- Learned that locals suffered fatigue, poor health due to anemia.
- Found distrust of Western medicines & low adoption of iron pills.

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2. Define

- *How might we deliver iron in a simple, affordable, and culturally acceptable way to fight anaemia?*

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3. Ideate

- Explored solutions:
  - Food fortification? → Too costly.
  - Iron cookware? → Not practical for all.
  - Iron blocks added to cooking pots? → Simple, reusable.

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4. Prototype

- First iron ingots (block-shaped) tested in cooking pots.
- Villagers rejected → looked like scrap metal.
- Redesigned shape to something **culturally meaningful** → a **fish (symbol of luck & prosperity in Cambodia)**.

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5. Test

- Distribution of **Lucky Iron Fish** to families.
- Acceptance increased drastically.

- Clinical trials showed a **significant rise in iron levels** & reduction in anaemia.
- Affordable, reusable for 5+ years, easy to use in daily cooking.

Outcomes & Impact

- Adopted in **multiple countries** beyond Cambodia.
- Millions benefited from reduced anaemia.
- Became a **social enterprise**, selling fish globally → profits fund free distribution to needy communities.
- Showcased **cultural empathy + iterative prototyping** as the key to success.

https://www.youtube.com/watch?v=iY0D-PIcgB4